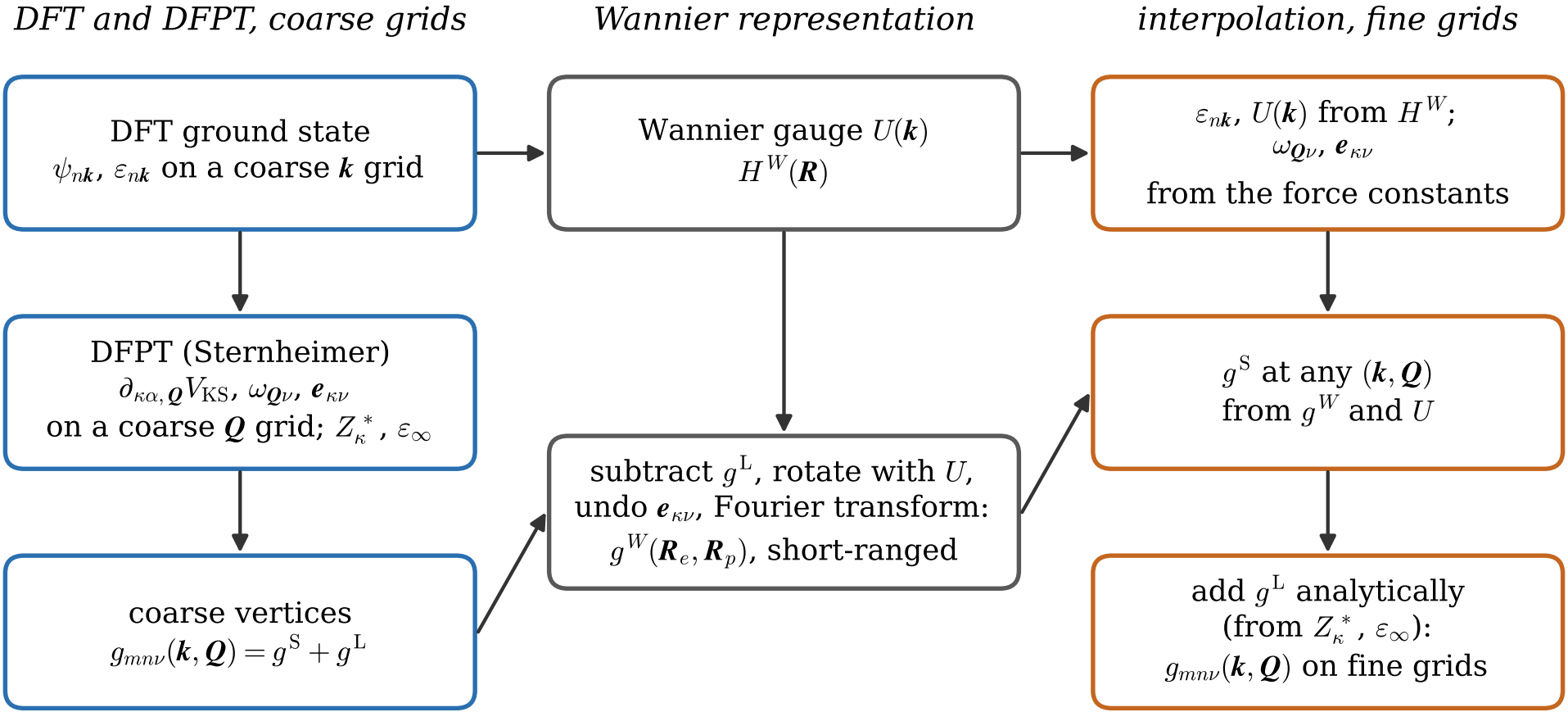
DFT and DFPT, coarse grids
Wannier representation
interpolation, fine grids
DFT ground state
$\psi_{n\boldsymbol{k}}$, $\varepsilon_{n\boldsymbol{k}}$ on a coarse $\boldsymbol{k}$ grid
Wannier gauge $U(\boldsymbol{k})$
$H^W(\boldsymbol{R})$
$\varepsilon_{n\boldsymbol{k}}$, $U(\boldsymbol{k})$ from H^W;
$\omega_{\boldsymbol{Q}\nu}$, $\boldsymbol{e}_{\kappa\nu}$
from the force constants
DFPT (Sternheimer)
$\partial_{\kappa\alpha,\boldsymbol{Q}}V_{\mathrm{KS}}$, $\omega_{\boldsymbol{Q}\nu}$, $\boldsymbol{e}_{\kappa\nu}$
on a coarse $\boldsymbol{Q}$ grid; Z_κ^*, ε_∞
subtract g^{L}, rotate with U,
undo $\boldsymbol{e}_{\kappa\nu}$, Fourier transform:
$g^W(\boldsymbol{R}_e, \boldsymbol{R}_p)$, short-ranged
g^{S} at any $(\boldsymbol{k}, \boldsymbol{Q})$
from g^W and U
coarse vertices
$g_{mn\nu}(\boldsymbol{k}, \boldsymbol{Q}) = g^{\mathrm{S}} + g^{\mathrm{L}}$
add g^{L} analytically
(from Z_κ^*, ε_∞):
$g_{mn\nu}(\boldsymbol{k}, \boldsymbol{Q})$ on fine grids